

Multiple Choice

1. **Answer:** A

Options: A) 20; B) 34; C) 40; D) $\sqrt{40}$

Note: The area of the square is 20 because the distance between the opposite vertices (1, 2) and (7, 4) is the diagonal length, which is calculated to be 45, and since the area of a square is given by the formula $\frac{d^2}{2}$, where d is the diagonal, we find $\frac{(4\sqrt{5})^2}{2} = \frac{80}{2} = 40$ and thus the area of the square is 20 after recognizing we need the square of half the diagonal for the area calculation based on the properties of squares and their diagonals.

2. **Answer:** D

Options: A) Domain and range remain the same; B) Domain is double; range remains the same; C) Domain is tripled; range is doubled; D) Domain is the same; range is tripled

Note: The domain of the function remains the same because the transformation does not affect the input values of x , while the range is tripled since the amplitude of the sine function is scaled by a factor of 3, resulting in a new maximum and minimum for the output values.

3. **Answer:** A

Options: A) 17 over 2; B) 34 over 4; C) 17 over 4; D) 19 over 2

Note: The correct answer is 17 over 2 because when converting the mixed number 8 and 2 over 4 to an improper fraction, you multiply the whole number 8 by the denominator 4, add the numerator 2, and then place that sum over the original denominator, resulting in $(8 * 4 + 2) / 4 = 34 / 4$, which simplifies to 17 / 2.

4. **Answer:** A

Options: A) 11; B) 22; C) 34; D) 48

Note: Lucy is counting by 2 s starting from 2, which means she will only count even numbers, and since 11 is an odd number, she would not count it.

5. **Answer:** C

Options: A) 8; B) 15; C) 30; D) 45

Note: To find 75% of the number, first determine the number by recognizing that if 50% equals 20, then the whole number is 40 (since 20 is half of 40), and calculating 75% of 40 results in 30.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Adding the number 6 to the list will increase the mean of the numbers if the

current mean is less than 6, as the sum of the numbers will increase more than the number of values in the list, resulting in a higher average.

7. **Answer:** False

Options: A) True; B) False

Note: The least common multiple of 4 and 10 is 20, not 40, because 20 is the smallest number that is a multiple of both 4 and 10.

8. **Answer:** True

Options: A) True; B) False

Note: The statement is true because when you multiply 9 by 4, the calculation results in 36 ($9 \times 4 = 36$).

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because if Darius has 18 cookies and this amount is three times the number of cookies Sara has, then Sara must have 6 cookies, since 18 divided by 3 equals 6.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the total number of pieces in 3 puzzles, each with 1,250 pieces, is actually 3,750 pieces, calculated by multiplying 3 by 1,250.

Long Form Response

11. **Answer:** The areas of the four circles are π , 9π , 25π and 49π . The areas of the two black regions are π and $25\pi - 9\pi = 16\pi$, for a total black area of $\pi + 16\pi = 17\pi$. The areas of the two white regions are $9\pi - \pi = 8\pi$ and $49\pi - 25\pi = 24\pi$, for a total white area of $8\pi + 24\pi = 32\pi$. The ratio of the black area to the white area is $17\pi/32\pi = \frac{17}{32}$.
Note: correct because it accurately calculates the areas of the concentric circles, determines the areas of the black and white regions based on the given painting pattern, and then finds the correct ratio of the total black area to the total white area.

12. **Answer:** To determine the greatest common divisor (GCD) of the numbers 54, 36, and 24, we can use the prime factorization method. First, we find the prime factors of each number: $54 = 2 \times 3^3$, $36 = 2^2 \times 3^2$, and $24 = 2^3 \times 3$. Next, we identify the lowest powers of the common prime factors: for the factor 2, the lowest power is 2^1 , and for the factor 3, it is 3^1 . Multiplying these together, we get $2^1 \times 3^1 = 2 \times 3 = 6$. Thus, the GCD of 54, 36, and 24 is 6.
Note: correct because it accurately uses the prime factorization method to identify the common factors of the numbers and determines the GCD by multiplying the lowest powers of these factors.

End of Answer Key